

Do Enlarged Perivascular Spaces Differ Across Four Study Groups?

An analysis of ePVS count, volume and diameter across four study groups

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1. Objective

Enlarged perivascular spaces (ePVS) are small fluid-filled spaces around brain blood vessels. They are used as a marker of small-vessel health and of how well the brain clears fluid [1,2]. The question here is whether three ePVS measures differ across the four study groups (A to D): the number of spaces, their total volume, and their mean diameter. I tested this after accounting for region size, age, sex, and handedness. The main analysis uses the whole white-matter mask, so each subject contributes one value. A regional breakdown is included as an exploratory follow-up.

2. Data and Quality Control

The dataset has 1,612 rows: 124 subjects, each measured in 13 white-matter masks (a whole mask, left and right hemispheres, and five lobar regions per hemisphere). I ran the usual checks before analysing anything (Table 1). There were no missing values, no duplicate subject-mask rows, and no negative numbers. Every subject had all 13 masks, and each subject's age, sex, handedness, and group stayed the same across their rows. As a sanity check on the mask hierarchy, the whole-brain count matched the sum of the left and right hemisphere counts. Modality only had one value (epc), so I did not use it as a covariate. Twenty-one rows (1.3%) had a count of zero. I read these as regions where no ePVS were found rather than errors, and kept all rows.

Table 1. Quality-control summary.

Check	Result
Rows	1612
Columns	11
Unique subjects	124
Unique masks	13
Missing values	0
Duplicate subject-mask rows	0
Negative numeric values	0
Modality (constant, not used as covariate)	epc
Age range (years)	7.38 to 17.20
All subjects have 13 masks	True
Demographics constant per subject	True
Whole-brain count = Left + Right	True
Zero-ePVS rows (no ePVS detected)	21 (1.3%)
Subjects per group	A=55; B=11; C=26; D=32

Table 2. Sample characteristics by group.

Group	N	Age, mean (SD)	Female, n (%)	Right-handed, n (%)	WM volume, mean (SD)
A	55	11.3 (2.0)	25 (45%)	49 (89%)	426760 (54653)
B	11	12.2 (1.6)	3 (27%)	10 (91%)	439008 (60253)
C	26	12.0 (2.3)	17 (65%)	26 (100%)	433267 (44883)
D	32	10.9 (2.1)	11 (34%)	26 (81%)	423251 (56196)

Values are mean (SD) or n (%). The sample is children and adolescents (age 7.4 to 17.2 years). The groups are unequal (A = 55, B = 11, C = 26, D = 32), and Group B is small. Age, sex, and handedness were only mildly unbalanced across groups ($p = 0.14, 0.07, \text{ and } 0.14$), but I still adjusted for them.

3. Methods

Outcome measures. Larger regions hold more ePVS just because there is more tissue, so raw counts and volumes are not comparable across subjects. In this data raw count tracked mask volume closely ($r = 0.83$). To get around this I used size-normalised outcomes: **ePVS density** (count per 10,000 voxels) and **volume fraction** (percent of the region that is ePVS). After normalising, the relationship with region size dropped to $r = 0.48$ (Figure 3), which is what we want. Mean **diameter** does not scale with region size, so I left it on its original scale. Diameter values of zero just mean no ePVS were found, so I set those to missing for the diameter model.

Transformations. Counts and volumes were right-skewed, so I log-transformed density and volume fraction before modelling. This brought them close to normal. Diameter was already roughly symmetric and was left alone (Figure 1).

Primary models. For each outcome I ran an ANCOVA (a linear regression): *outcome ~ group + age + sex + handedness*. Group is the predictor I care about, and age, sex, and handedness are the covariates. I used Type II ANOVA because it copes well with unequal group sizes, and I report partial η^2 as the effect size. I also ran a plain one-way ANOVA and a Kruskal–Wallis test for comparison. The Kruskal–Wallis test does not make a normality assumption, but it also cannot adjust for covariates, so I treat it as a backup rather than the main result. I checked the ANCOVA assumptions with a Shapiro–Wilk test on the residuals and Levene’s test for equal variances.

Multiple comparisons. I corrected the three main group p-values (density, volume fraction, diameter) using the Benjamini–Hochberg false-discovery-rate (FDR) method. I went with FDR rather than Bonferroni because the three outcomes are related and Bonferroni would be too strict here, costing power for little gain [3]. When the corrected group effect was significant, I followed up with **Tukey HSD** pairwise tests and reported **Cohen’s d**.

Secondary (exploratory) regional analysis. I combined the left and right lobar masks into five bilateral regions (frontal, parietal, temporal, occipital, insula) and re-ran the density ANCOVA in each one, then FDR-corrected across the five regions. I am treating these regional results as exploratory.

4. Results

Distributions. ePVS counts and volumes were right-skewed but looked close to normal after the log transform. Diameter was roughly symmetric already (Figure 1). **Group differences.** ePVS **density differed significantly across the groups** (ANCOVA $p = 0.016$, FDR $p = 0.049$, partial $\eta^2 = 0.084$, which is a medium effect). The plain one-way ANOVA ($p = 0.004$) and the Kruskal–Wallis test ($p = 0.014$) pointed the same way. Volume fraction ($p = 0.42$) and diameter ($p = 0.82$) did not differ across groups (Figure 2, Table 3). The model assumptions held up (residual normality $p = 0.11$, equal variances $p = 0.96$).

Table 3. ePVS measures by group (whole white matter).

Outcome	A	B	C	D
ePVS count	388.96 (86.22)	475.64 (127.00)	412.81 (92.99)	393.47 (84.29)
ePVS density (/10k vox)	9.06 (1.19)	10.70 (1.64)	9.44 (1.33)	9.23 (1.09)
ePVS volume fraction (%)	0.47 (0.11)	0.55 (0.10)	0.50 (0.13)	0.50 (0.12)
ePVS mean diameter (vox)	1.99 (0.09)	1.99 (0.07)	2.00 (0.09)	2.00 (0.10)

Values are mean (SD). Group B shows the highest count and density; diameter is almost identical across groups.

Post-hoc (density). The Tukey HSD tests put **Group B at the top**. Its density was higher than Group A ($p = 0.002$, Cohen’s $d = 1.19$) and higher than Group D ($p = 0.014$, $d = 1.10$). Groups A, C, and D were close to each other. **Covariates.** Density went up with age ($r = 0.44$, $p < 0.001$, Figure 5), which makes sense in a growing cohort. Sex and handedness barely moved density. **Regional (exploratory).** The group difference in density showed up mainly in the **frontal** region (ANCOVA $p = 0.009$, FDR $p = 0.046$). The other four regions were not significant (Figure 4).

5. Discussion

The group difference showed up in the **number** of ePVS per unit of tissue (density) and not in their volume fraction or diameter. Group B had the highest density, and this was strongest in the frontal region. A higher count with no change in diameter fits a picture of more perivascular spaces rather than bigger ones [2]. The age relationship lines up with what we would expect, since perivascular spaces tend to become more visible as children grow, and it is a good reason to keep age in the model. Taken together, this is a fairly modest signal: the groups differ on one outcome (density) and mostly in one region (frontal).

6. Strengths, Limitations, and Future Directions

Strengths. The analysis normalises for region size, adjusts for age, sex, and handedness, and pairs the ANCOVA with a non-parametric test so the main result does not hang on one set of assumptions. It corrects for multiple testing, reports effect sizes next to the p-values, and runs end to end from a single script with a fixed seed, so every number is reproducible.

Limitations. The biggest caveat is that Group B has only 11 subjects. Its density is high and the effect size is large, but that still rests on very few people, so it really needs to be checked in a bigger sample. The group labels and imaging details are anonymised, so I cannot say what the groups actually represent. ePVS numbers also depend on how the images were processed [4], and there was no scan-quality or motion information to rule out technical differences between groups. The data are from a single time-point, so nothing can be said about change over time.

Future directions and alternative approaches. The clearest next step is a larger sample, especially for Group B, and ideally follow-up scans so ePVS can be tracked over time rather than at a single time-point. On the modelling side, a mixed-effects model could keep the left and right hemispheres as separate observations, a negative-binomial count model could work on the raw counts with region volume as an offset, and matching or weighting the groups on age and sex would be another way to handle confounding. Since this is a developmental cohort, and the lab studies language and dyslexia with multimodal imaging, a natural extension is to relate ePVS to diffusion-MRI white-matter integrity and language-network connectivity, and to follow ePVS across development and link it to reading and cognitive outcomes. Consistent with the lab's machine-learning work on predicting cognitive and language decline, ePVS metrics could also feed predictive models built from structural, diffusion and functional MRI rather than being tested one measure at a time. Finally, automated deep-learning segmentation of ePVS would reduce measurement noise and make it practical to scale to larger, multi-site samples.

7. Conclusion

The primary analysis evaluated whether whole-white-matter ePVS density, volume fraction, and mean diameter differed across groups after adjusting for age, sex, and handedness. Regional analyses were considered exploratory. Findings involving Group B should be interpreted cautiously due to its small sample size.

8. Reproducibility Statement

The entire analysis is a single Python script (*analysis/epvs_analysis.py*) run with a fixed random seed. It uses relative paths, prints package versions, and regenerates every table, figure, and the results file (*results.json*) from the raw CSV. To reproduce:

- `pip install -r requirements.txt`
- `python analysis/epvs_analysis.py`

References

- [1] Wardlaw JM, Benveniste H, Nedergaard M, et al. Perivascular spaces in the brain: anatomy, physiology and pathology. *Nature Reviews Neurology*. 2020;16:137–153.
- [2] Wardlaw JM, Smith EE, Biessels GJ, et al. Neuroimaging standards for research into small vessel disease (STRIVE). *Lancet Neurology*. 2013;12(8):822–838.
- [3] Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J R Stat Soc B*. 1995;57(1):289–300.
- [4] Sepehrband F, Barisano G, Sheikh-Bahaei N, et al. Image processing approaches to enhance perivascular space visibility and quantification using MRI. *NeuroImage*. 2019;197:243–254.

Appendix: Figures

Figure 1. Distributions of ePVS outcomes (n = 124)

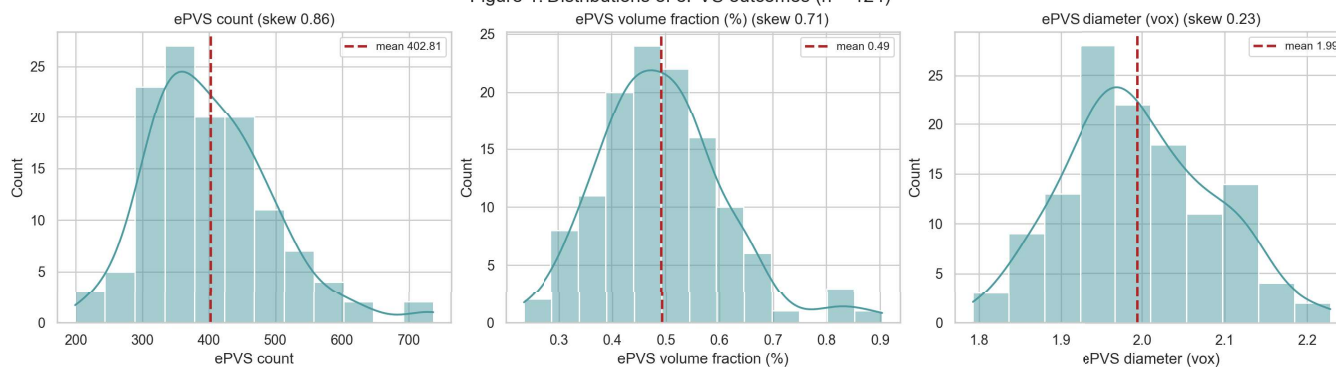


Figure 1. Distributions of the ePVS outcomes. Counts and volumes are right-skewed (log-transformed for testing); diameter is roughly normal.

Figure 2. ePVS outcomes by study group

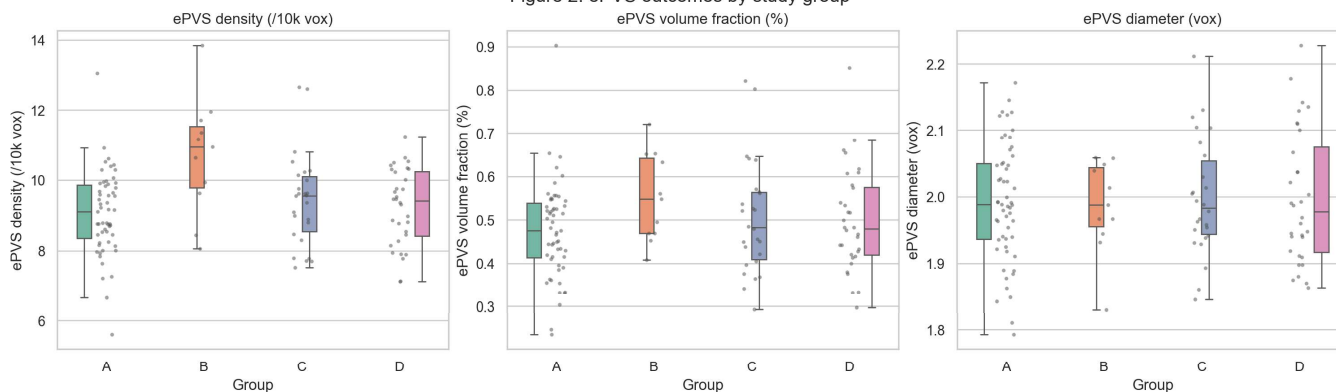


Figure 2. ePVS outcomes by group. Group B has the highest density; volume fraction and diameter look similar across groups.

Figure 3. Why mask size must be accounted for

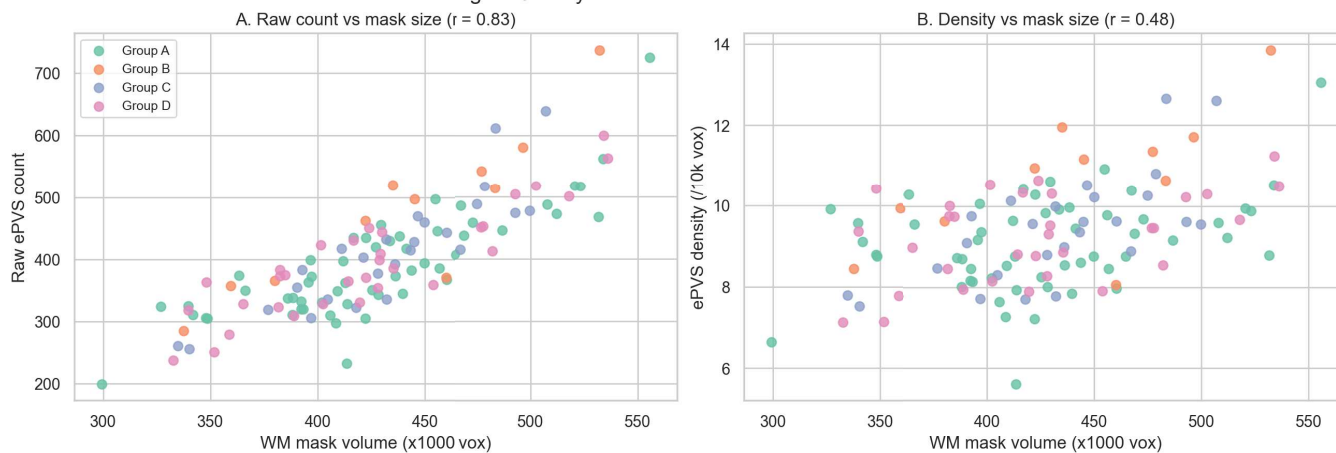


Figure 3. Why mask size matters. (A) Raw ePVS count rises with mask volume; (B) density is nearly flat, so it is a fair, size-free measure.

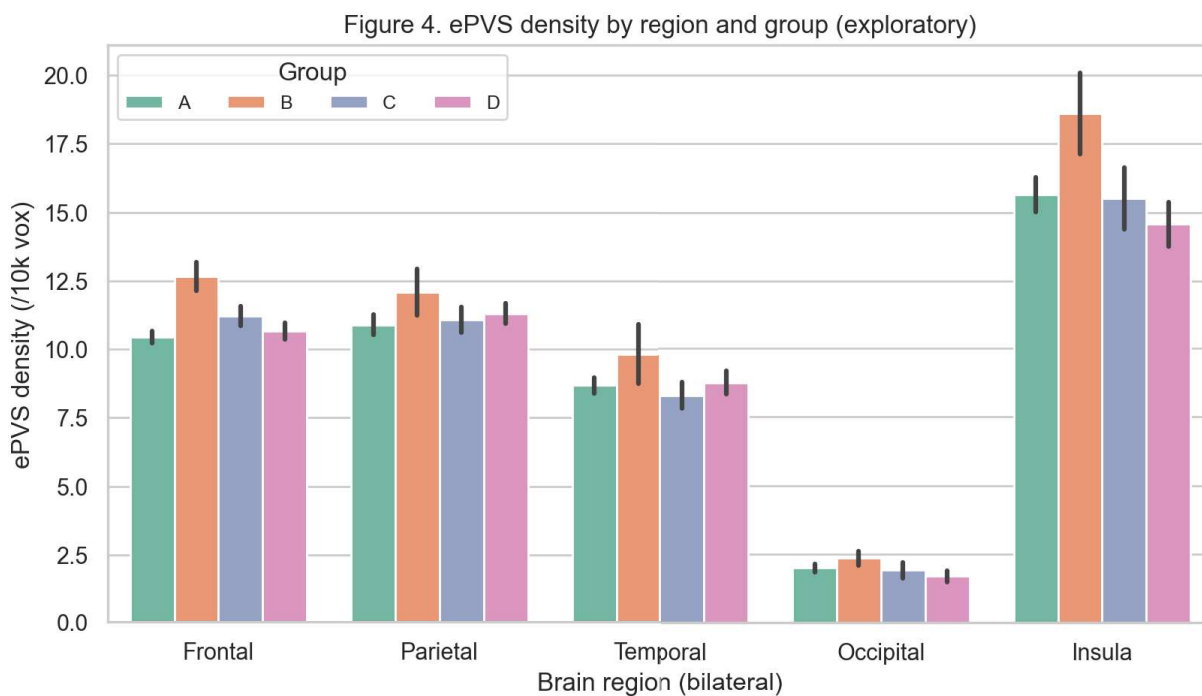


Figure 4. ePVS density by bilateral region and group (exploratory). Group B is highest in the frontal region.

